

A Model for Boundary Diffusion Controlled Creep in Polycrystalline Materials

R. L. Coble

Citation: *J. Appl. Phys.* **34**, 1679 (1963); doi: 10.1063/1.1702656

View online: <http://dx.doi.org/10.1063/1.1702656>

View Table of Contents: <http://jap.aip.org/resource/1/JAPIAU/v34/i6>

Published by the [American Institute of Physics](#).

Additional information on J. Appl. Phys.

Journal Homepage: <http://jap.aip.org/>

Journal Information: http://jap.aip.org/about/about_the_journal

Top downloads: http://jap.aip.org/features/most_downloaded

Information for Authors: <http://jap.aip.org/authors>

ADVERTISEMENT



AIPAdvances

Special Topic Section:
PHYSICS OF CANCER

Why cancer? Why physics? [View Articles Now](#)

VII. DISCUSSION

The results of the iterative emission phase-angle distributions have shown that the intuitive picture of the multipacting process is essentially correct. The important assumption, $v/v_0 = \text{constant}$, has been shown to have a theoretical basis in the sense that the multipacting electron distribution with respect to $k = v/v_0$ is rather narrow and shows a peak near the semi-empirically determined value $k=3$ on the minimum break-

down field strength curve. Furthermore, it has been demonstrated that both the emission and arrival phase-angle distributions are of essentially the same form.

ACKNOWLEDGMENT

The authors would like to express their gratitude to Dr. R. M. Conkling for his invaluable assistance in the preparation of the computer program used in this work.

A Model for Boundary Diffusion Controlled Creep in Polycrystalline Materials

R. L. COBLE

*Ceramics Division, Department of Metallurgy, Massachusetts Institute of Technology,
Cambridge 39, Massachusetts*

(Received 15 October 1962)

The creep rate ($\dot{\epsilon}$) predicted by the boundary diffusion (D_b) model is

$$\dot{\epsilon} \approx 150\sigma D_b W \Omega / (GS)^3 kT,$$

where σ is the stress, W is the boundary width, (GS) is the average grain size, and Ω is vacancy volume. The stress dependence is the same as the lattice diffusion model, given by C. Herring, while the grain size dependence and the numerical constant are greater for boundary diffusion. Discussion of the mechanism of creep in polycrystalline alumina is based on the differences between the lattice and boundary diffusion models.

INTRODUCTION

THE macroscopic shape change of crystals by diffusional transport of atoms was shown to be possible by Nabarro¹ and Herring.² Herring's analysis gave an explicit expression for the creep rate of polycrystalline materials at high temperatures:

$$\dot{\epsilon} = 10\sigma D_l \Omega / (GS)^2 kT, \quad (1)$$

where $\dot{\epsilon}$ is the strain rate, σ is the applied stress, D_l is the lattice diffusion coefficient, Ω is the vacancy volume, GS is the grain size, k is Boltzmann's constant, and T is temperature.

Polycrystalline Al_2O_3 has been found to satisfy this diffusional creep model for wide ranges of stress and temperature. Folweiler's,³ Warshaw's,⁴ and Beauchamp, Baker, and Gibbs⁵ measurements on polycrystalline alumina show this behavior from 1100° to 1800°C at stresses from 200 to 25 000 psi and grain sizes ranging from 5–50 μ . The relations between strain rate, stress, grain size, and temperature also conform to the bulk

diffusion model for polycrystalline BeO ^{6,7} over a similar broad range of experimental variables. Further, diffusion coefficients calculated from creep data^{3–7} and measured directly for Al in Al_2O_3 ⁸ and Be in BeO ⁹ are within the estimated error limits for these measurements.

In Al_2O_3 , the aluminum ion diffusion coefficient⁸ is larger than the oxygen ion diffusion coefficient¹⁰ by several orders of magnitude. When the creep rate is limited by lattice atomic diffusion, the least mobile species present (O^{2-}) would be expected to fix the rate of this process. Paladino and Coble¹¹ proposed that the deformation process in aluminum consists of Al diffusion through the lattice and oxygen diffusion along the grain boundaries where its diffusion coefficient is enhanced in comparison with the values in the lattice.¹⁰ Similar enhancement of anion diffusion at grain boundaries has been observed in alkali halides.¹²

This raises the question, "If boundary diffusion of

¹ F. R. N. Nabarro, *Report of a Conference on the Strength of Solids* (The Physical Society, London, 1948), p. 75.

² C. Herring, *J. Appl. Phys.* **21**, 437 (1950).

³ R. Folweiler, *J. Appl. Phys.* **32**, 773 (1961).

⁴ S. I. Warshaw and F. H. Norton, *J. Am. Ceram. Soc.* **45**, 479 (1962).

⁵ E. K. Beauchamp, G. S. Baker, and P. Gibbs, "Impurity Dependence of Creep of Al_2O_3 ," University of Utah, WADD Contract No. AF 33(616)-6832, Project No. 0(7-7350) (1961).

⁶ R. Chang, *J. Nucl. Mater.* **2**, 174 (1959).

⁷ B. Chandler, E. C. Duderstadt, and J. White, submitted to *J. Nucl. Mater.*

⁸ A. E. Paladino, Jr., and W. D. Kingery, *J. Chem. Phys.* **37**, 957 (1962).

⁹ S. B. Austerman, U. S. Atomic Energy Commission Report NAA-SR-3170, (1958).

¹⁰ Y. Oishi and W. D. Kingery, *J. Chem. Phys.* **33**, 480 (1960).

¹¹ A. E. Paladino and R. L. Coble, *J. Am. Ceram. Soc.* (to be published).

¹² J. F. Laurent and J. Benard, *J. Phys. Chem. Solids* **7**, 218 (1958).

oxygen is required, may not the boundary diffusion process control the over-all creep rate?" Without knowing whether the stress and grain size dependences of creep are different for lattice and boundary diffusion models, the qualitative confirmation of the lattice diffusion model is not conclusive. In this paper a model for boundary diffusion controlled creep in polycrystalline materials is derived in order to test the uniqueness of the lattice diffusion model.

GRAIN SHAPE

In a polycrystalline matrix the grain shape and orientation varies from grain-to-grain such that selection of a suitable single analog for property analysis is difficult. A spherical analog is selected here in order that the analysis be directly comparable to that of Herring for lattice diffusion.

DRIVING FORCE

The driving force for diffusional creep was shown by Nabarro¹ and Herring² to result from a change in equilibrium vacancy concentration with stress. Their result can be reduced to a simple expression for the change in vacancy concentration

$$\Delta C = C_0 \sigma \Omega / kT, \quad (2)$$

where C_0 is the equilibrium vacancy concentration at temperature T in a stress-free crystal, and σ is the component of the local stress normal to the boundary. Upon application of a uniaxial tensile (+) stress the maximum change in vacancy concentration (ΔC) occurs on boundaries perpendicular to the stress axis, where vacancy creation produces lattice expansion in the direction parallel to the stress allowing the applied stress to perform work. On boundaries parallel to the stress, lattice expansion due to vacancy creation occurs in a direction where the resolved stress and work performed are zero. Therefore, the equilibrium vacancy concentration on boundaries parallel to the stress is C_0 from Eq. (2) and is only temperature-dependent.

For a spherical grain the stress direction defines a polar axis; the vacancy concentration at the equator is C_0 , and the vacancy concentration at the poles changes by the amount calculated from Eq. (2). We assume in the discussion that follows that a tensile stress is applied and that excess vacancies created at the pole diffuse to the equator. Since each half of the sphere acts independently, we need only consider one hemisphere; the diffusion flux is zero at the pole and at the equator.

VACANCY SOURCE

In order to define the vacancy source on the spherical surface, we may first consider the characteristics of a polycrystalline array which are to be approximated. The grains have different numbers of faces, but each face is nearly planar and, therefore, presents a constant angle

with respect to the stress axis. Each face has a uniform normal tensile component from which an equilibrium change in vacancy concentration is calculable from Eq. (2). If no diffusion occurs, the vacancy concentrations would be uniform on each face, and would change abruptly at the three grain edges in passing from face-to-face.

The vacancy creation and annihilation rates govern the local geometric change for which there are two restrictions. The space filling polycrystalline matrix must continue to fill space, and the stress distribution must not change continuously. One acceptable geometric change occurs if each boundary maintains parallelism with its former orientation as the grains are deformed. This requires that a given crystal face have a uniform vacancy generation rate over the whole face.

For the spherical analog, we can select suitable areas for the vacancy source and sink, and assume that the generation and annihilation rates are uniform. For a first approximation we assume that the rates are not only uniform, but also equal. (If they were assumed to be unequal, as would be appropriate for an oriented matrix of anisometric grains, the predicted rate would differ by a numerical factor with no effect on the stress or grain size dependence.)

With the restriction that the sphere maintains constant volume, the area of the vacancy source must be equal to the area of the vacancy sink. Since the boundary of rotational symmetry between equal areas on a hemisphere lies at 60° below the pole, the problem is to evaluate the concentration gradient at the 60° boundary. At this point the flux is a maximum relative to the average gradient given by $(\Delta C / (R\pi/2))$, where R is the sphere radius.

For steady-state creep, Fick's first law applies and the flux at the 60° boundary is

$$J \text{ vac sec}^{-1} = D_v N [\Delta C / (R\pi/2)] (W) 2\pi R \sin 60, \quad (3)$$

where D_v is the diffusion coefficient of vacancies in the boundary, N is a proportionality constant relating the average vacancy gradient $\Delta C / (R\pi/2)$ and the maximum gradient $1/R(dC/d\theta)_{\theta=60}$, W is the effective boundary width and $(2\pi R \sin 60)$ is the length of the (zone) where the diffusion flux is a maximum. Thus the cross-sectional area for diffusion is $2\pi RW \sin 60$.

The problem reduces to an evaluation of N and its sensitivity to the vacancy source and vacancy sink distribution.

UNIFORM SOURCE AND SINK

In the spherical case with the above assumptions the flux increases linearly with area (A) away from the pole:

$$dJ = BdA = B2\pi R^2 \sin\theta d\theta,$$

and

$$J_\theta = B2\pi R^2 (1 - \cos\theta), \quad (4)$$

where B is the number of vacancies created per unit area per sec. Equation (4) applies at $60 \geq \theta \geq 0$. (Then $J_{\max} = J_{\theta=60} = B\pi R^2$). Similarly from $60 \leq \theta \leq 90$,

$$J = B2\pi R^2 \cos\theta. \quad (5)$$

Using Fick's first law, the gradient at the surface $dC/Rd\theta$ is proportional to the flux, and inversely proportional to the cross-sectional area ($2\pi RW \sin\theta$), which gives for $60 \leq \theta \leq 90$

$$dC/Rd\theta = B2\pi R^2(1 - \cos\theta)/D_v W 2\pi R \sin\theta, \quad (6)$$

and for $90 \geq \theta \geq 60$,

$$dC/Rd\theta = B2\pi R^2 \cos\theta/D_v W 2\pi R \sin\theta. \quad (7)$$

The average gradient is

$$\frac{\Delta C}{R\pi/2} = \frac{BR}{DW\pi/2} \left[\int_0^{60} \frac{(1 - \cos\theta)d\theta}{\sin\theta} + \int_{60}^{90} \frac{\cos\theta d\theta}{\sin\theta} \right] \quad (8)$$

and the maximum gradient from either Eq. (6) or (7) is

$$(dC/Rd\theta)_{\theta=60} = 0.577BR/D_v W. \quad (9)$$

The value of N in Eq. (3), given by division of Eq. (9) by (8) is

$$N = \frac{(dC/Rd\theta)_{\theta=60}}{\frac{\Delta C}{R\pi/2}} = \frac{0.577BR/D_v W}{\frac{BR}{D_v W\pi/2} \left[\int_0^{60} \frac{(1 - \cos\theta)d\theta}{\sin\theta} + \int_{60}^{90} \frac{\cos\theta d\theta}{\sin\theta} \right]}, \quad (10)$$

and

$$N = \frac{0.577\pi}{2\{-\ln(1 + \cos\theta)\}_0^{60} + [\ln \sin\theta]_{60}^{90}} = 2.15.$$

NONUNIFORM SOURCE AND SINK

Because of the fact that a stress gradient is assumed to exist, the assumption that vacancies are created uniformly over the source may not be valid. A trial solution for distributed source [$B = \phi(\theta)$] with an arbitrary continuous variation in flux was determined by requiring that the flux be zero at the pole ($\theta = 0$) and at the equator ($\theta = \pi/2$) and be a maximum at $\theta = 60^\circ$. The expression $J = B2\pi R^2 \theta^n \sin 2\theta$ satisfies these criteria, with $n = 1.2$. The gradient then becomes

$$dC/Rd\theta = BR\theta^{1.2} \sin 2\theta/D_v W \sin\theta = (BR/D_v W)\theta^{1.2} \cos\theta.$$

This function was integrated numerically. The result, following the procedure above, is: $N = 1.52$.¹³

From these analyses, it is apparent that N is of the

¹³ Another solution, by D. W. Readey, who assumed that the vacancy concentration varied with the resolved normal stress component, gave $N = 1.43$.

order of unity. A maximum value occurs with a cusped maximum for the flux such as results from the assumption of a uniform source ($N = 2.15$). Thus, for a boundary diffusion model, the diffusion flux is relatively insensitive to the uniformity of the vacancy source and sink; the diffusion flux given by Eq. (3) with $N = 2.15$ becomes

$$J = 7.4D_v W \Delta C \text{ vacancies/cm sec.} \quad (11)$$

CREEP MODEL

With the above flux of vacancies diffusing from the uniform source of area πR^2 the volume change is: $\pi R^2 dR/dt$,

$$J a_0^3 = \pi R^2 (dR/dt) = 7.4D_v W \Delta C a_0^3,$$

and

$$\dot{\epsilon} = 1/R (dR/dt) = 7.4D_v W \Delta C a_0^3 / \pi R^3. \quad (12)$$

Substituting $\Delta C = C_0 \sigma a_0^3 / kT$ and ($D_b = D_v C_0 a_0^3$), where D_b is the diffusion coefficient for atoms in the boundary gives

$$\dot{\epsilon} = 7.4 \sigma D_b W a_0^3 / \pi R^3 kT. \quad (13)$$

In addition to the shape change by diffusion, the creep rate may be enhanced by shear stress relaxation at grain boundaries. Herring² has shown that $\dot{\epsilon}_{(\text{relaxed})} = \frac{5}{2} \dot{\epsilon}_{(\text{unrelaxed})}$. By the nature of the correction the same factor applies for boundary diffusion. With the above factor and calling the grain size (GS) = $2R$, the creep rate is

$$\dot{\epsilon} = 148 \sigma (D_b W) a_0^3 / (GS)^3 kT, \quad (14)$$

which includes all the factors required for direct comparison with the lattice diffusion model [Eq. (1)].

DISCUSSION

The stress dependence of creep rate is found to be the same for both the lattice and boundary diffusion models. Therefore, the only experimental variable for distinguishing between the models is the grain size dependence. Average values were taken from Folweiler's plotted data for Al_2O_3 .³ [An evaluation of n for $\sigma/(GS)^n$ is not warranted because the grain sizes were not measured on individual samples.] For each temperature the values of $\sigma/(GS)^2$ should be constant for perfect agreement with the lattice diffusion model. Though the values are not constant, the agreement is better than that which results with a $\sigma/(GS)^3$ correlation. Therefore, for Al_2O_3 the data confirm the lattice diffusion model.

Paladino and Coble's assumption¹¹ that diffusion of oxygen at grain boundaries is sufficiently enhanced such that lattice diffusion of aluminum controls the net transport of material is supported by the above result. The Al^{3+} diffusion coefficient measurements have been conducted on polycrystalline specimens of constant grain size. Whether Al diffusion is enhanced at grain bound-

aries cannot be determined from the available data. There is quantitative agreement between measured diffusion coefficients and those calculated from creep data using the lattice diffusion model.

The boundary width (W) introduces a degree of arbitrariness to the boundary diffusion model because only the product $D(W)$ has been separable from the diffusion measurements. From earlier measurements, the width has been assumed to be several interatomic spacings. However, recent measurements of diffusion coefficients of oxygen in alumina¹⁰ and for Cl^- in polycrystalline NaCl ¹² and as affected by dislocation density¹⁴ have shown that the thickness of the region within which the diffusion coefficient is enhanced may be of the order of 10^2 to 10^4 Å rather than the 5 Å or so which seemed appropriate for metals.

If the grain size is of the order of thickness within which diffusion is enhanced, Eq. (3) should be more appropriate for boundary diffusion; for larger grain sizes, Eq. (14) should apply. The grain sizes of the

¹⁴ L. W. Barr, I. M. Hoodless, J. A. Morrison, and R. Rudham, *Trans. Faraday Soc.* **56**, 449, 697 (1960).

alumina samples on which creep measurements were conducted ranged from $5\text{--}50\text{ }\mu$,³⁻⁵ thus being an order of magnitude larger than the apparent widths of boundaries deduced from diffusion measurements. Since the boundary diffusion model is rejected by the observed grain size dependence, and agreement within $\sim 4\times$ exists between the measured aluminum ion diffusion coefficients and those calculated from creep data with Eq. (1), it appears that the applicable bulk diffusion model refers to lattice (rather than enhanced boundary) diffusion. Measurements of creep rates in samples of larger grain sizes provide a more decisive test of this conclusion; a transition to creep rate control by boundary diffusion should occur at larger grain sizes.

ACKNOWLEDGMENTS

The author gratefully acknowledges helpful discussions with W. D. Kingery, D. W. Readey, and D. R. Chipman.

This research was supported by the U. S. Atomic Energy Commission under its Contract AT(30-1)-2574.

Linewidth and Temperature Shift of the R Lines in Ruby

D. E. MCCUMBER AND M. D. STURGE

Bell Telephone Laboratories, Incorporated, Murray Hill, New Jersey

(Received 29 November 1962; in final form 28 January 1963)

Improved experimental data have been obtained between 20° and 350°K for the widths and temperature shifts of the R lines in ruby. Above 77°K the results can be accurately described in terms of Raman scattering of Debye-model phonons. There is no evidence for measurable contributions to the widths and shifts from direct processes involving emission or absorption of a single resonant phonon. Below 77°K the principal contributions are from crystal inhomogeneities.

WE have measured the temperature dependence of the position and width of the R lines in the optical spectrum of dilute ruby and have compared it with the predictions of a simple model. The temperature dependence in the range 77° to 400°K is consistent with a two-phonon Raman-process model^{1,2} having a single characteristic (Debye) temperature. Below 77°K the linewidth is primarily due to inhomogeneous crystal strains^{2,3} whose contributions are assumed to be independent of temperature. Consequently, it is not possible to determine optically which phonon processes are important at low temperatures. Our results are con-

sistent with (but more accurate than) previous data.⁴⁻⁶ The data of reference 4 extends the range of agreement with theory up to 700°K .

The R lines were measured in absorption to avoid the reabsorption errors inherent in fluorescence measurements. They were measured in σ and π polarizations on two specimens of Linde ruby containing, respectively, 1.8×10^{18} and 2.2×10^{19} Cr^{3+} ions/cc. The specimens were mounted in a variable-temperature cell in the entrance optics of a Jarrell-Ash 1.8-m scanning spectrometer, used with a spectral slit width of 0.056 cm^{-1} . Linewidths of the different components and polarizations of a given R line agreed and were re-

¹ J. H. Van Vleck, *Phys. Rev.* **57**, 426 (1940).

² A. Kiel, *Phys. Rev.* **126**, 1292 (1962), has considered the width of the R lines, but as has been pointed out by R. H. Silsbee [*Phys. Rev.* **128**, 1726 (1962)] his analysis is inapplicable in several important respects.

³ A. L. Schawlow, *J. Opt. Soc. Am.* **51**, 472 (1961); A. L. Schawlow, A. H. Piksis, and S. Sugano, *Phys. Rev.* **122**, 1469 (1961).

⁴ K. S. Gibson, *Phys. Rev.* **8**, 38 (1916).

⁵ O. Deutschbein, *Ann. Phys.* **14**, 712 (1932); **14**, 729 (1932); **20**, 828 (1934).

⁶ A. L. Schawlow, *Advances in Quantum Electronics*, edited by J. R. Singer (Columbia University Press, New York, 1961), p. 50.